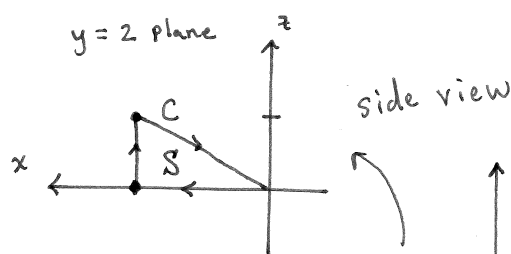


HW 11 Stokes' Thm

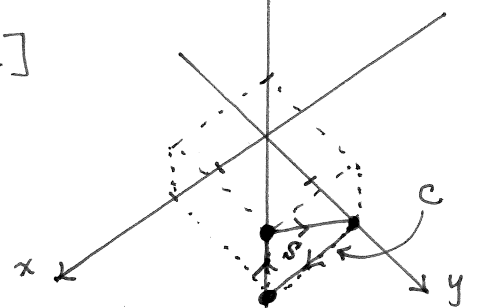


Problem 1: $\vec{F} = [2y, 3z, x]$

$$(a) \quad \text{curl}(\vec{F}) = [0-3, 0-1, 0-2] = [-3, -1, -2]$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n} \, dA$$

$$= \iint_S 1 \, dA = \text{Area}(S) = \frac{1}{2}(2)(1) = 1$$



with this orientation, The Normal must point toward the x-axis

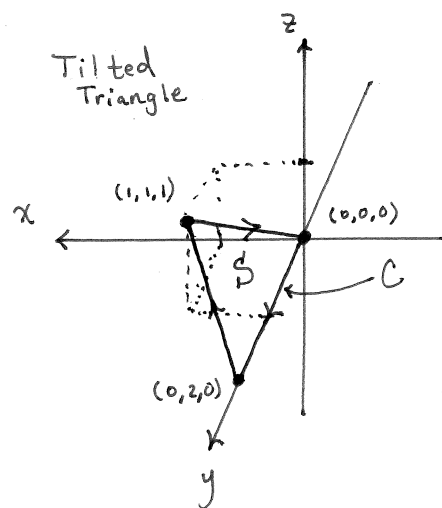
$$\Rightarrow \vec{n} = [0, -1, 0]$$

(unit normal)

$$(b) \quad \text{curl}(\vec{F}) = [-3, -1, -2]$$

$$\text{Note: } \text{Area}(S) = \frac{1}{2} \|[2, 0, -2]\|$$

$$= \frac{1}{2} \sqrt{8} = \sqrt{2}$$



with this orientation, the normal should point toward the xy-plane

To find a normal to the triangular plane we compute a cross-product:

$$[0, 2, 0] \times [1, 1, 1]$$

$$= [2, 0, -2]$$

$$\Rightarrow \vec{n} = \left[\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2} \right]$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n} \, dA$$

$$= \iint_S -\frac{3\sqrt{2}}{2} + \frac{2\sqrt{2}}{2} \, dA$$

$$= \iint_S -\frac{\sqrt{2}}{2} \, dA$$

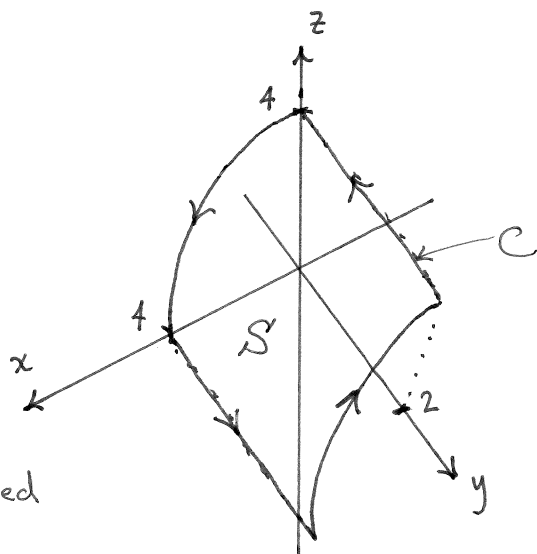
$$= -\frac{\sqrt{2}}{2} \text{Area}(S) = -\frac{\sqrt{2}}{2}(\sqrt{2})$$

$$= -1$$

$$(c) \quad \vec{F} = [y, 2-3yz, x^2+y^2]$$

$$\begin{aligned} \text{curl}(\vec{F}) &= [2y+3y, 0-2x, 0-1] \\ &= [5y, -2x, -1] \end{aligned}$$

Since the unit normal $\vec{n}$ and $\text{curl}(\vec{F})$ are not constant, we will need to parameterize S and compute



$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}(\vec{F})(\vec{r}) \cdot \vec{N} \, du \, dv = (*)$$

Since x & z have a circular relationship, consider cylindrical coord.
Note: $r=4$ @ every pt of S

$$\begin{array}{lll} x = 4 \cos \theta & u = y & x = 4 \cos v \\ y = y & \longrightarrow & y = u \\ z = 4 \sin \theta & v = \theta & z = 4 \sin v \end{array} \quad \rightarrow \quad \vec{r}(u,v) = [4 \cos v, u, 4 \sin v]$$

$0 \leq u \leq 2 \quad 0 \leq v \leq \frac{\pi}{2}$
this defines R

$$\vec{r}_u = [0, 1, 0]$$

$$\vec{r}_v = [-4 \sin v, 0, 4 \cos v]$$

$$\Rightarrow \quad \vec{N} = [4 \cos v, 0, 4 \sin v]$$

A quick check ($v=0$) shows $\vec{N}$ is pointing in the correct direction.

$$\text{curl} \vec{F}(\vec{r}) = [5u, -8 \cos v, -1]$$

$$\text{curl} \vec{F}(\vec{r}) \cdot \vec{N} = 20u \cos v - 4 \sin v$$

$$\begin{aligned} (*) &= \iint_R 20u \cos v - 4 \sin v \, du \, dv = \int_0^{\frac{\pi}{2}} \int_0^2 20u \cos v - 4 \sin v \, du \, dv \\ &= \int_0^{\frac{\pi}{2}} 10u^2 \cos v - 4u \sin v \Big|_0^2 \, dv \\ &= \int_0^{\frac{\pi}{2}} 40 \cos v - 8 \sin v \, dv \\ &= 40 \sin v + 8 \cos v \Big|_0^{\frac{\pi}{2}} \\ &= (40+0) - (0+8) = 32 \end{aligned}$$